

Exam Date & Time: 14-Jun-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VI Semester Makeup Examination, June -2025

INFORMATION RETRIEVAL [IT-4409]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Apply Boolean retrieval to process the query (machine AND learning) OR (deep AND networks) using an inverted index. Show step-by-step execution. (5)
 - A)
 - B) Apply the step-by-step process of constructing an inverted index for the following document collection:
 - Doc1: "Machine learning is powerful." (5)
 - Doc2: "Deep learning improves AI."
- 2) Analyze the effectiveness of Front Coding in dictionary compression. Given the words: {apple, application, apply, banana, band, banner}, apply Front Coding and discuss its advantages and limitations. (5)
 - A)
 - B) Show the effectiveness of VSM compared to Probabilistic Retrieval Models in real-world search applications. (5)
- 3) Interpret the significance of Mean Average Precision (MAP) in evaluating ranked retrieval systems. (5)
 - A)
 - B) Discuss the Rocchio Algorithm for query refinement. (5)
- 4) (5)
 - A)

Given the query "*Neural Networks for Medical Diagnosis*" and the following document-term matrix:

Document	Neural	Networks	Medical	Diagnosis
D1	1	1	1	0
D2	1	1	0	0
D3	1	1	1	1

1. Compute the relevance score (RSV) for each document using the probabilistic model, given:

- o Document frequency (*df*): Neural (3), Networks (3), Medical (2), Diagnosis (1).
- o Total documents (*N*) = 3.

Apply the required interpretation to rank the documents and justify why D3 is the most relevant in medical IR.

- B) Discuss the search user experience in the case of an e-commerce website. (5)
- 5) A news aggregation platform is building a specialized crawler to collect articles from 50,000 news websites. The platform faces the following challenges:
- A)
- 30% of articles are duplicated (e.g., syndicated content, republished press releases).
 - Dynamic loading (e.g., JavaScript-rendered content) causes the crawler to miss key text.
 - Rate-limiting by news sites leads to incomplete data collection.
 - Server overload occurs when crawling large multimedia articles (videos, high-res images). (5)

Implement a crawling strategy that:

1. Minimizes duplicate content storage.
2. Handles dynamic content effectively.
3. Avoids being blocked by news websites.

- B) Describe how to minimize unnecessary crawling without compromising content coverage. (5)

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